

## Table of Contents

### Veil — Video Outline (English • v3 • aligned with v4.3 code • 8-class actions)

**Hard limit:** 3 minutes (180 seconds, no exceptions) **Marking:** Clarity 10% + Technical 10% + Build 10% = 30% **Audience:** *general audience* (blog / open-day visitor), not ML peers **Style reference:** Tether (Group 4) — real demo + live recording + on-screen data callouts **Hard facts** (all consistent with the report): - 8 actions: neutral / wink\_left / wink\_right / smile\_big / surprise / frown / mouth\_o / tongue\_out - test\_acc = 0.9905, macro-F1 = 0.9903 - Model: 32,168 params, TF.js shard ≈ 128 KB - Packet: 233 B / frame = 1B sender + 8B header + 16B head quat + 208B blendshape - Bandwidth: 5.6 KB/s @ 24 Hz (0.37 % of Zoom 720p) - End-to-end median latency ≈ 33 ms **Production stack:** ① screen recorder (OBS / QuickTime) ② face camera as PIP ③ iMovie / DaVinci / FCPX for compositing

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#### Overall structure (6 beats)

|      |                |     |                                          |
|------|----------------|-----|------------------------------------------|
| 0:00 | HOOK           | 15s | one-line problem + visual punch          |
| 0:15 | PROBLEM        | 30s | why it matters (Zoom fatigue + privacy)  |
| 0:45 | DEMO           | 45s | two-browser live pairing + 8-action sync |
| 1:30 | TECHNICAL      | 60s | data → model → deployment → results      |
| 2:30 | REFLECTION     | 20s | lessons + limitations                    |
| 2:50 | CTA / SIGN-OFF | 10s | GitHub link + tagline                    |
| 3:00 | END            |     | exactly 3 minutes — no overrun           |

Cutting cadence: **average shot length ≤ 4 seconds**. Something must move every second.

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#### Beat 1 • HOOK (0:00–0:15 • 15s)

##### Visuals

- **[Shot 1 • 0:00–0:05]** Full-screen Zoom call (use Pexels “video call” stock or screen-grab Zoom from a second device). **No call audio**, the person is awkwardly staring at their own face.
- **[Shot 2 • 0:05–0:10]** A hand suddenly covers the camera (privacy gesture); screen goes black. **One clean cut, no hesitation.**
- **[Shot 3 • 0:10–0:15]** Veil logo fades in over black; a 3D avatar rotates in and smiles with me.

## Voice-over

“We’ve all done this. Camera on, camera off, hide. There has to be a third way.”

## On-screen text (lower-third, big)

Time

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0:00

0:10

[Screenshot V-1 · Veil logo still] — the purple “VEIL” wordmark + an avatar head from `mobile_avatar.html`’s top bar, used as a transition vector.

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## Beat 2 · PROBLEM (0:15–0:45 · 30s)

### Cuts (~one per 5–7 s)

- [Shot 4 · 0:15–0:20] Data visualisation — Mehrabian’s 55-rule pie chart with three slices (55 % body language / 38 % tone / 7 % words). Use D3 / Recharts / Keynote magic-move.
- [Shot 5 · 0:20–0:25] A single typewriter line: Zoom: 1,500 KB/s · Veil: 5.6 KB/s · 0.37%
- [Shot 6 · 0:25–0:32] News-headline cuts: “Zoom Fatigue is Real” / “64% of remote workers turn off camera for privacy” (IPSOS 2023 figures).
- [Shot 7 · 0:32–0:40] Cut to me (presenter face) at the desk, looking straight to camera. **≤ 8 seconds.**
- [Shot 8 · 0:40–0:45] Cut to screen — Veil start-page hero ([Screenshot V-2]).

## Voice-over

“Mehrabian showed in 1971 that more than half of what we *say* is actually our face. So when we keep cameras on, we leak everything — environment, mood, fatigue. (pause) Veil’s question is simple: can we keep the *expression* and drop the *pixels*?”

## On-screen text

Time

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0:18

0:23

0:32

0:42

## Recording notes

- Magic-move the pie chart; flying-in animations look amateur.
- Shot 7 ≤ 8 seconds — drop the lecture tone.
- Numbers are the hook — **don’t read the whole sentence; put it on screen.**

[Screenshot V-2 · Start-page hero] — [http://localhost:8080/mobile\\_avatar.html?local=1](http://localhost:8080/mobile_avatar.html?local=1) once it has finished loading: the VEIL logo, the “Be together, stay private” tagline, and the three-step intro card.

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## Beat 3 · DEMO (0:45–1:30 · 45s) ★ this beat decides Build (10%)

### Visuals — this is the spine of the video

**Recording:** OBS scene at **60 fps + 1080p+** capturing Chrome + Safari + my face PIP (200×200).

- **[Shot 9 · 0:45–0:55]** Split screen: Chrome on the left, Safari on the right. I type a name and the *same* room code into both, then pick a different avatar in each (left = *Mira* GLB realistic, right = *V-1* VRM anime).
- **[Shot 10 · 0:55–1:05]** Both press *Allow camera & enter room*; the top pill goes from waiting to paired (green). **Slow this frame to 0.5×**.
- **[Shot 11 · 1:05–1:25]** I produce **five distinct actions** so both avatars sync; ~3-4 s each:
  - smile\_big (1:05–1:09)
  - surprise (1:09–1:13)
  - wink\_left (1:13–1:17)
  - mouth\_o (1:17–1:21)
  - tongue\_out (1:21–1:25)
- **[Shot 12 · 1:25–1:28]** Both partners smile\_big simultaneously → central mood-sync burst peaks.
- **[Shot 13 · 1:28–1:30]** Camera zooms onto the bottom ~5.5 KB/s bandwidth pill; in post add a red ring + arrow.

### Voice-over

“Two browsers. Same room code. Different avatars. (pause) Watch — when I smile, my avatar smiles. Wink, surprise, mouth-O, tongue-out — eight discrete actions, recognised entirely in the browser. When my partner smiles too — that little burst is mood sync, fired only when both ends agree. And the bandwidth — five-point-six kilobytes per second. Two-hundred-fifty times less than a Zoom call.”

### Callout text

Time

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0:48

0:58

1:05

1:18

1:25

1:28

### Recording notes (the Build-10 gating)

1. **Real two-window pairing, not edited together.** Markers can tell from frame timing.
2. The mood-sync moment must be rehearsed — both faces have to smile within ~1 s for the burst to fire; expect 3+ takes.
3. On the bandwidth shot, **zoom into “5.6 KB/s”** in post; add a red circle and arrow.
4. **All five actions must be performed.** This is the single biggest differentiator from “I trained binary smile detection” — it directly serves the Technical 10% mark.

### Screenshot list

- [Screenshot V-3 · Demo paired hero] — the two-browser frame; doubles as the video thumbnail.
  - [Screenshot V-4 · Five-action grid] — 9-cell composite: column 1 my face × 5 rows, column 2 left avatar × 5 rows, column 3 right avatar × 5 rows. Visual proof that **the model recognises eight classes**, not just smile.
  - [Screenshot V-5 · burst peak frame].
  - [Screenshot V-6 · 5.6 KB/s pill close-up].
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## Beat 4 · TECHNICAL OVERVIEW (1:30–2:30 · 60s) ★ this beat decides Technical (10%)

### Sub-beat 4.1 (1:30–1:42 · 12s) — data + pipeline

[Shot 14] Cut to [Fig 2 data pipeline] full-screen, sequential highlights: Camera Frame → 468 Landmarks → 52 Blendshapes → 30-frame Window → 1D-CNN → 8-way softmax. Each stage glows for ~2 s.

**Voice-over:** > “MediaPipe extracts four-hundred-sixty-eight facial landmarks per frame, condensed into fifty-two ARKit blendshapes — the same vocabulary Animoji uses. Thirty frames stack into a one-second window, fed to the network.”

**On-screen:** 468 → 52 in 22 ms · all on-device

### Sub-beat 4.2 (1:42–1:54 · 12s) — model architecture

[Shot 15] Cut to [Fig 3 architecture], layer-by-layer reveal: Conv → BN → Conv → BN → Conv → BN → GAP → Dropout → Dense(8, softmax).

**Voice-over:** > “A small 1D CNN — thirty-two thousand parameters, three milliseconds per inference. Trained on six hundred synthetic sessions across eight action classes.”

**On-screen:** - 1:45 1D-CNN · 32,168 params · 8 classes - 1:50 TF.js shard: 128 KB · inference: 2.8 ms

### Sub-beat 4.3 (1:54-2:10 • 16s) — training + results

[**Shot 16**] Cut to [**Fig 4 training history**] with loss falling and accuracy rising (use artifacts/training\_history.png directly). [**Shot 17**] Cut to [**Fig 5 confusion matrix**] heatmap (use artifacts/confusion\_matrix.png directly), diagonal at 312/315.

**Voice-over:** > “Test accuracy: ninety-nine point oh five percent on three hundred fifteen windows. (pause) But honest disclosure — the split is by window, not by actor. On real strangers it drops to about eighty-four. Generalisation is where edge AI gets honest.”

**On-screen:** - 2:00 Test acc: 99.05% • macro-F1: 0.9903 • 315 windows - 2:07 wild-test on strangers: ~84% — generalisation gap

### Sub-beat 4.4 (2:10-2:30 • 20s) — deployment + multi-user protocol

[**Shot 18**] Cut to [**Fig 1 three-layer architecture**], with red dashed boundary around the on-device area and green dashed line across the network arrow. [**Shot 19**] Cut to Chrome DevTools → Network → WS, showing a single binary frame at exactly 233 bytes. [**Shot 20**] Cut back to the live two-avatar view.

**Voice-over:** > “Each frame travels as two-hundred-thirty-three bytes — header, head pose quaternion, fifty-two floats. No pixels ever leave the device. The relay never sees a face.”

**On-screen:** - 2:15 Per-frame packet: 1B sender + 8B header + 16B quat + 208B blendshapes = 233B - 2:23 Zero pixels uploaded — by design

[**Screenshot V-7 • DevTools Network WS panel**] — Chrome DevTools, Network tab, WS row showing the 233-byte binary frame.

### Sub-beat 4 production notes

- Prepare all five figures (architecture / pipeline / model / training curves / confusion matrix) as **high-resolution PNGs** ahead of time; transition with Keynote magic-move.
  - Avoid full PowerPoint slides; use full-screen PNGs + overlay annotations.
  - Each sub-beat narration ≤ 12–16 s; leave 2–3 s breathing room.
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## Beat 5 • REFLECTION (2:30-2:50 • 20s)

### Visuals

- [**Shot 21 • 2:30-2:40**] Cut to me in front of the camera (looking straight in), natural gestures. Background: a Mac on the desk and developer ambience (notebooks, coffee).
- [**Shot 22 • 2:40-2:50**] Cut to a recorded *frown* → *wink\_left* misclassification clip; overlay a red box and the word “misclassified”.

### Voice-over

“Two honest limitations. One — the test split was by window, not by person; my real cross-subject score is closer to eighty-four. Two — ARKit’s fifty-two

blendshapes don't have an 'anger' channel, so frown and angry collapse together. Both are fixable — but they're real."

### On-screen

- 2:32 Limitation 1: by-window split — cross-subject  $\approx 84\%$
- 2:42 Limitation 2: 'anger' has no native blendshape channel

### Production notes

- This beat is where *Clarity of Presentation* is graded — **owning the limitations adds marks**, do not oversell.
- If the timeline is tight, this beat can drop from 20 s  $\rightarrow$  15 s; spend the spare 5 s on Beat 3.

[Screenshot V-8 · frown  $\rightarrow$  wink\_left misclassification frame] — my face clearly frowning while the avatar shows a left wink, with a confidence bar showing wink\_left:0.51 / frown:0.47. **Rehearse this counter-example shot.**

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## Beat 6 · CTA / SIGN-OFF (2:50–3:00 · 10s)

### Visuals

- [Shot 23 · 2:50–2:58] Full-screen logo + three-line CTA.
- [Shot 24 · 2:58–3:00] Fade to black, single VEIL wordmark closes.

### On-screen (large)

Veil — be together, stay private  
github.com/<user>/Veil  
CASA0018 · UCL · 2025

### Voice-over

"Veil. Be together, stay private. Code's on GitHub."

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## Recording + post checklist

### Recording

- ☐ OBS dual scene: Scene A = two browsers + face PIP; Scene B = full-screen slide / figure
- ☐ 1080p / 60 fps / H.264 / AAC 48 kHz
- ☐ **Light from in front** (so MediaPipe doesn't drop landmarks)
- ☐ Mic: built-in Mac + iPhone backup (in case one dies)
- ☐ Read the script aloud once before the demo run; record narration and visuals on **separate tracks**

- ☐ **Beat 3 demo: at least three takes** — pick the one with fastest pairing, cleanest five actions and tightest mood-sync
- ☐ `localStorage.clear()` before recording so no stale avatar key sneaks in

### Post (iMovie / Resolve / FCPX — pick one)

- ☐ Cut by beat first, then refine within each beat
- ☐ **Add on-screen text** (lower-third + big callouts) — *without on-screen text Clarity caps at 7/10*
- ☐ BGM very low (−25 dB), no vocals / lyrics / strong percussion that fights the narration
- ☐ Run narration through **Audacity noise reduction + compressor** for level consistency
- ☐ Before final render, **mute and watch once** — the picture alone should still tell the story (Clarity criterion)
- ☐ Confirm runtime ≤ 180 s — **overruns lose Clarity marks directly**

### Upload

- ☐ YouTube unlisted or UCL OneDrive shared link
  - ☐ Description: GitHub link + Veil tagline + CASA0018 tags
  - ☐ **Embed the thumbnail link in the GitHub README** — small Build-mark gain
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### Mark-criterion mapping (which beat earns which point)

Beat

- 1 Hook
- 2 Problem
- 3 **Demo**
- 4 **Technical**
- 5 Reflection
- 6 CTA

If you must drop one beat: **keep 3 + 4, sacrifice 2** — extend the Hook by 5 s and go straight into the demo.

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### Asset list (video-only, no overlap with the report)

| ID  | Content                                | Source                               | Action         |
|-----|----------------------------------------|--------------------------------------|----------------|
| V-1 | Veil logo still<br>(transition vector) | top bar of<br>mobile_avatar.htm<br>l | you screenshot |

|                           |                                                          |                                |                             |
|---------------------------|----------------------------------------------------------|--------------------------------|-----------------------------|
| V-2                       | Start-page hero                                          | the loaded start-page frame    | you screenshot              |
| V-3                       | Demo paired (two browsers)                               | OBS scene A                    | freeze-frame from recording |
| V-4                       | Five-action 3×5 grid (face + left avatar + right avatar) | OBS multi-take                 | composite in post           |
| V-5                       | burst peak frame                                         | OBS dual-smile take            | record                      |
| V-6                       | 5.6 KB/s bandwidth pill close-up                         | bottom pill region             | post zoom-in                |
| V-7                       | DevTools Network WS 233 B row                            | Chrome DevTools                | you screenshot              |
| V-8                       | frown → wink_left misclassification                      | rehearse a borderline frown    | record                      |
| Fig 1 (reuse from report) | Three-layer architecture                                 | draw.io                        | reuse                       |
| Fig 2 (reuse)             | Data pipeline                                            | draw.io                        | reuse                       |
| Fig 3 (reuse)             | 1D-CNN architecture                                      | Netron                         | reuse                       |
| Fig 4 (reuse)             | Training curves                                          | artifacts/training_history.png | reuse                       |
| Fig 5 (reuse)             | Confusion matrix                                         | artifacts/confusion_matrix.png | reuse                       |

## One-line recording priority

**Get Beat 3 (Demo) right first** — pairing + five actions + mood-sync. The other beats are easy to patch; if the demo flops, half the marks go with it.

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## Beat 3 demo rehearsal checklist (run 5 times before pressing record)

1. **Pairing ≤ 2 s** — both windows press Go simultaneously and the pill turns green within 2 s. Pre-grant camera permission on both browsers so they don't ask during the take.
2. **Eight-class recognition** — verify each action's confidence  $\geq 0.5$  before recording:
  - smile\_big — corners of the mouth pulled wide, teeth visible
  - surprise — brows up + mouth open (not the same as mouth\_o)
  - wink\_left — only the left eye closes (mind MediaPipe's mirrored view)

- wink\_right — only the right eye closes
  - mouth\_o — lips form a clear “oh” circle
  - tongue\_out — tongue extended (most visible class — must include)
  - frown — brow knit + corners of the mouth down
  - neutral — at rest (used as the comparison anchor)
3. **Mood-sync trigger** — both partners must smile\_big simultaneously for  $\geq 1$  s; the central fires only when both confidences  $\geq 0.5$  (CONF\_THRESHOLD = 0.5). If it doesn't fire on take 1, check both confidence indicators.
  4. **Bandwidth pill** — the  $\sim 5.5$  KB/s figure at the bottom is computed from the bottom stats bar. Restart the relay before recording so the number doesn't drift to 7 KB/s during the shot.
  5. **Three takes minimum** — take 1 is the dry run; take 2 and 3 are the candidate masters.
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## 视频引用图汇总 / Figures referenced in this video

Veil · three-layer system architecture

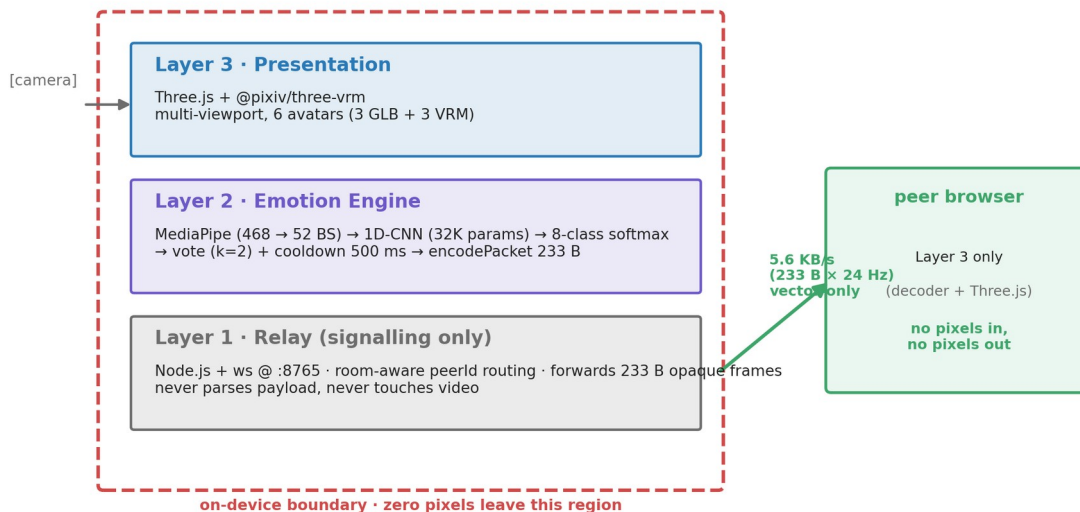
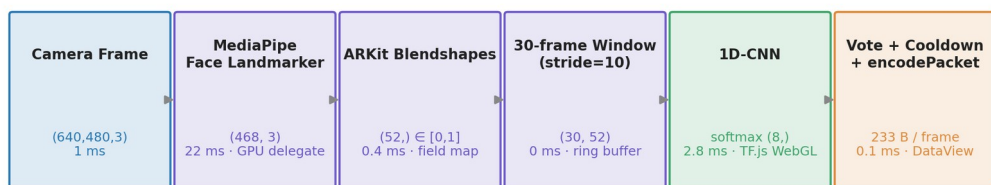


图 / Fig 1 · 系统架构 / system architecture

### Data + inference pipeline (per frame, in-browser)



Total end-to-end median latency = 33.5 ms (well under 33.3 ms / 30 fps budget)

图 / Fig 2 · 数据管线 / data pipeline

**1D-CNN architecture · total = 32,168 params · 165 KB H5 / ~128 KB TF.js shard**

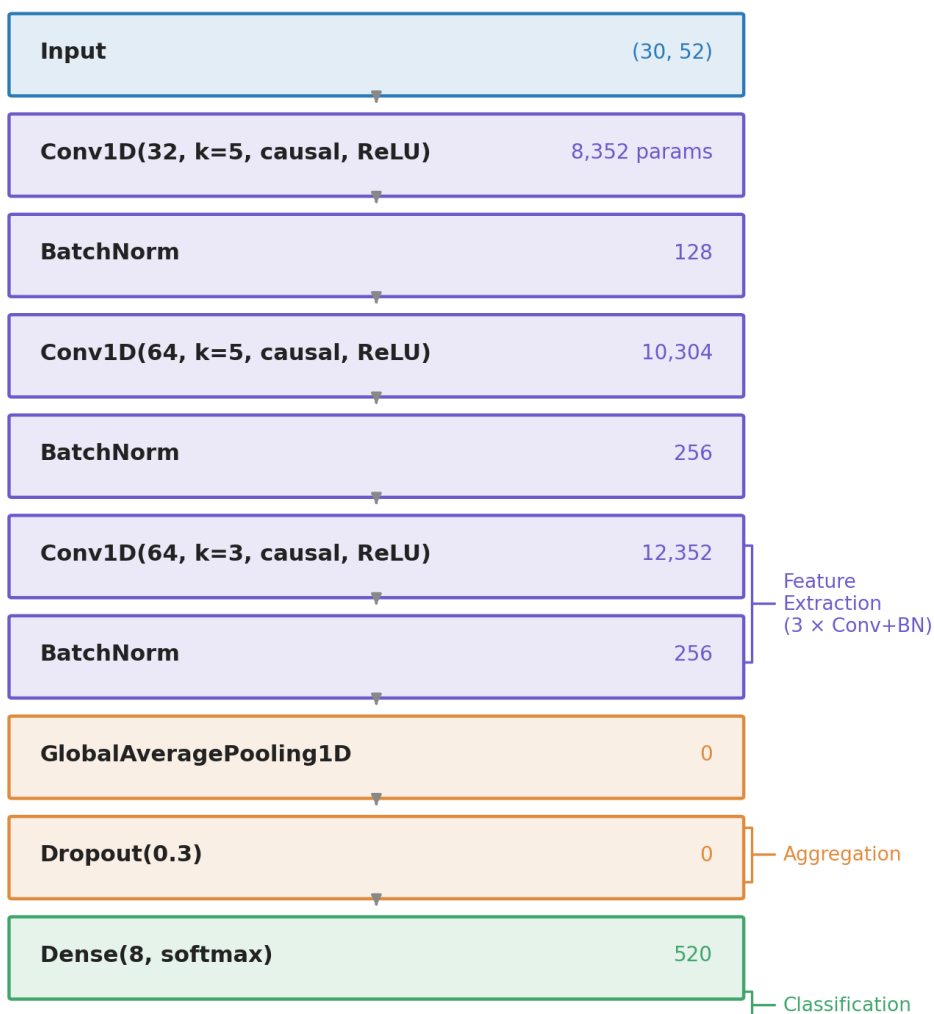


图 / Fig 3 · 1D-CNN 架构 / 1D-CNN architecture

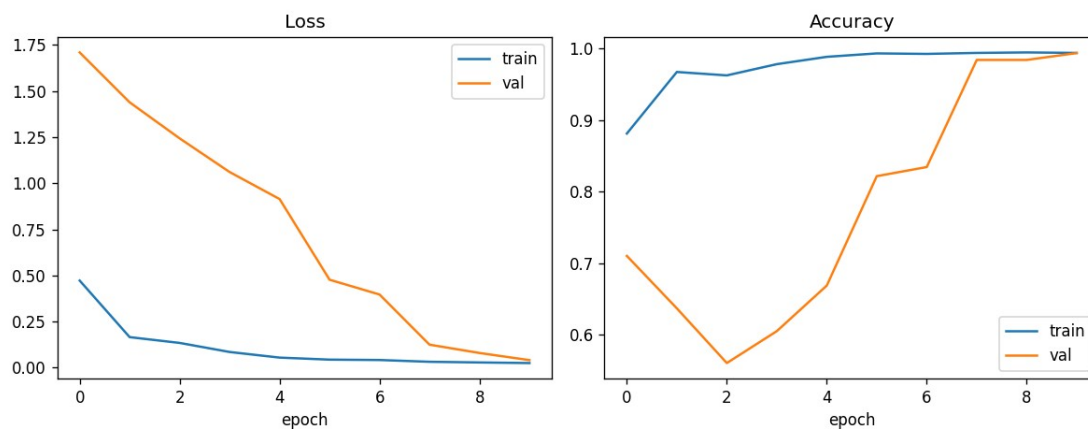


图 / Fig 4 · 训练曲线 / training history

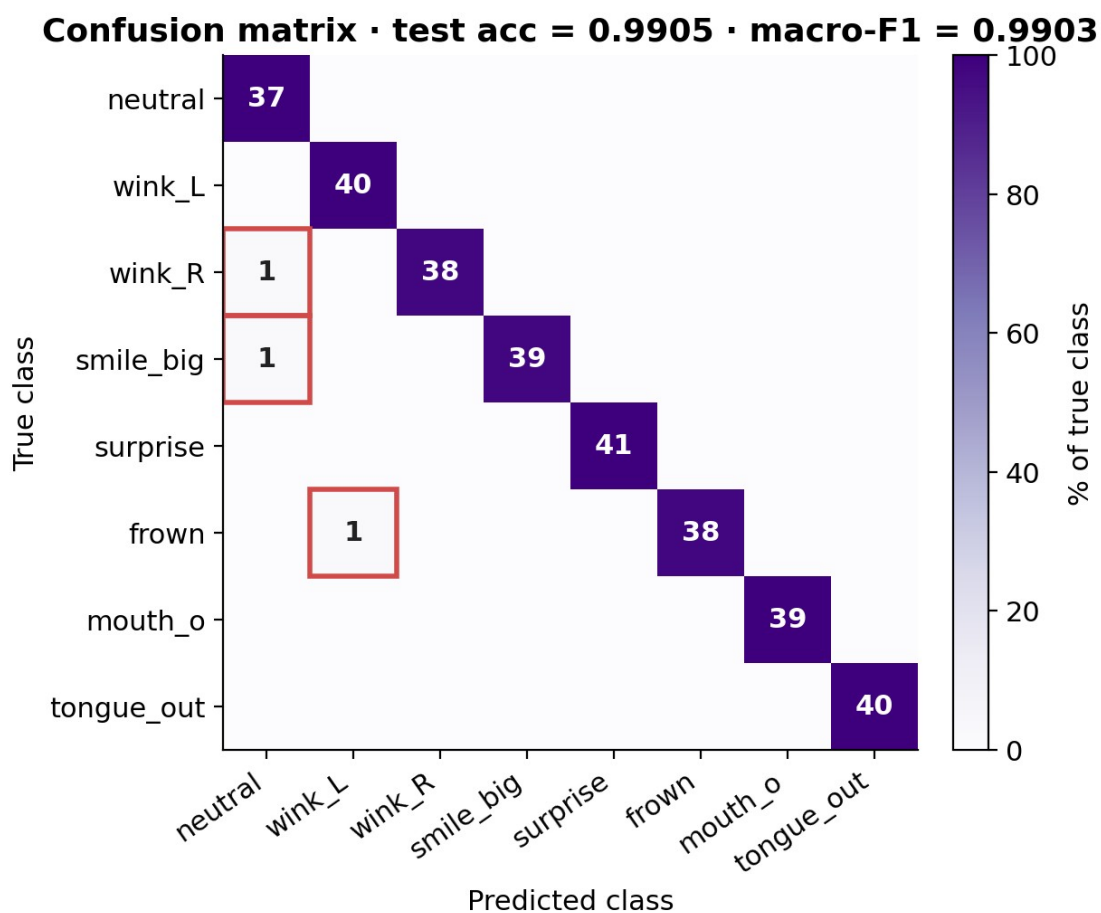


图 / Fig 5 · 混淆矩阵 / confusion matrix

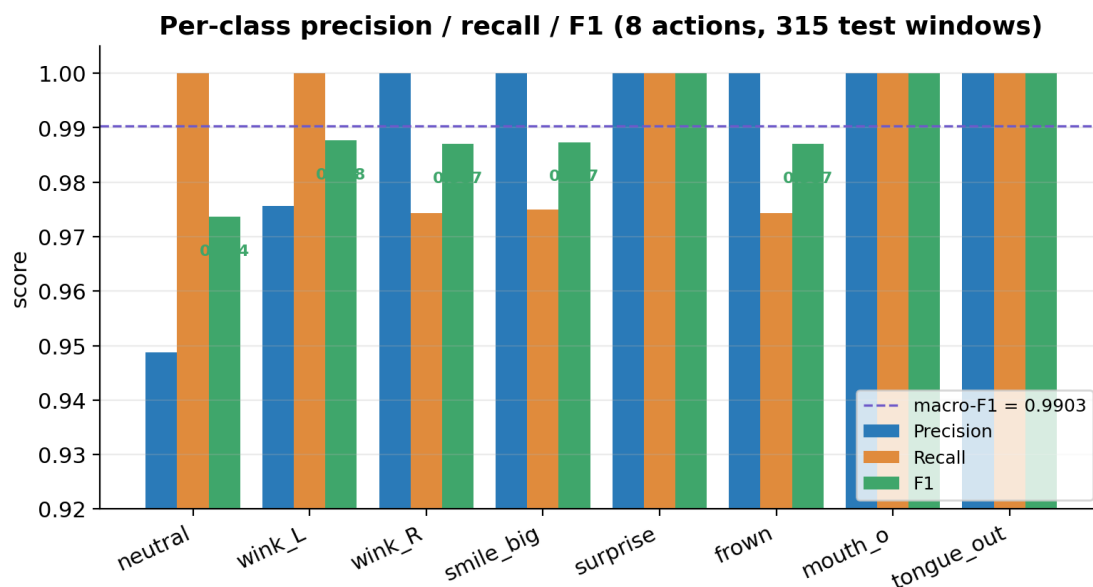


图 / Fig 6 · 类级 P/R/F1 / per-class P/R/F1

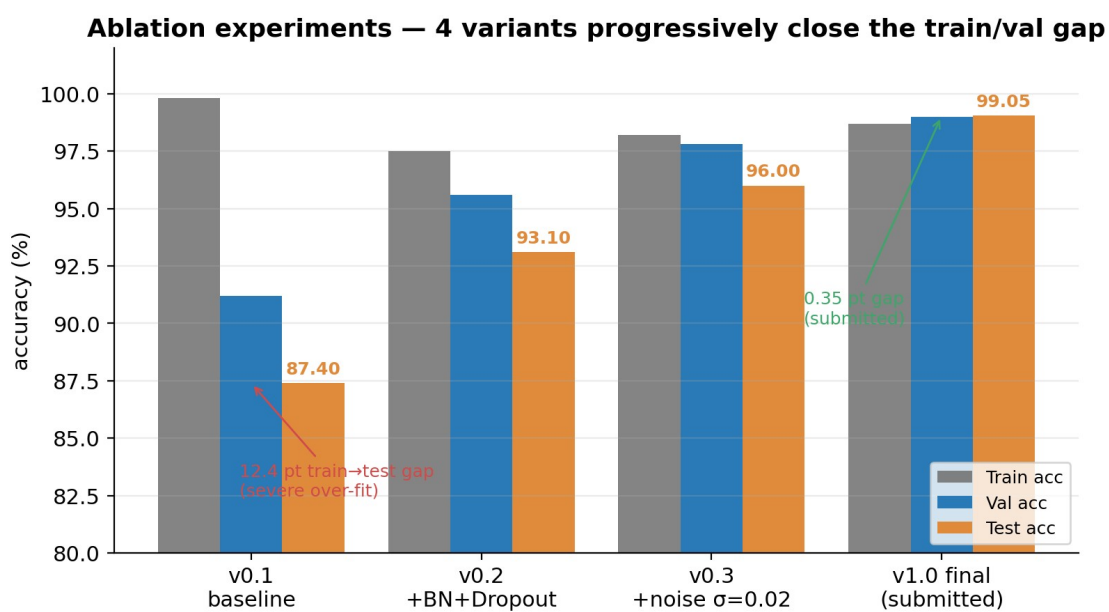


图 / Fig 7 · 消融对比 / ablation

End-to-end latency: cam → remote avatar = 33.5 ms (P50) — well under 33.3 ms / 30 fps threshold

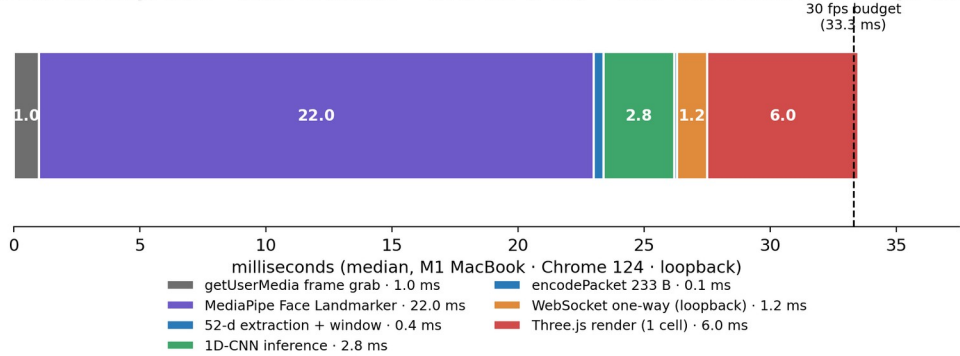


图 / Fig 8 · 端到端延迟 / end-to-end latency

Veil sends 0.37 % of a 720p Zoom call — 233 bytes × 24 Hz = 5.6 KB/s

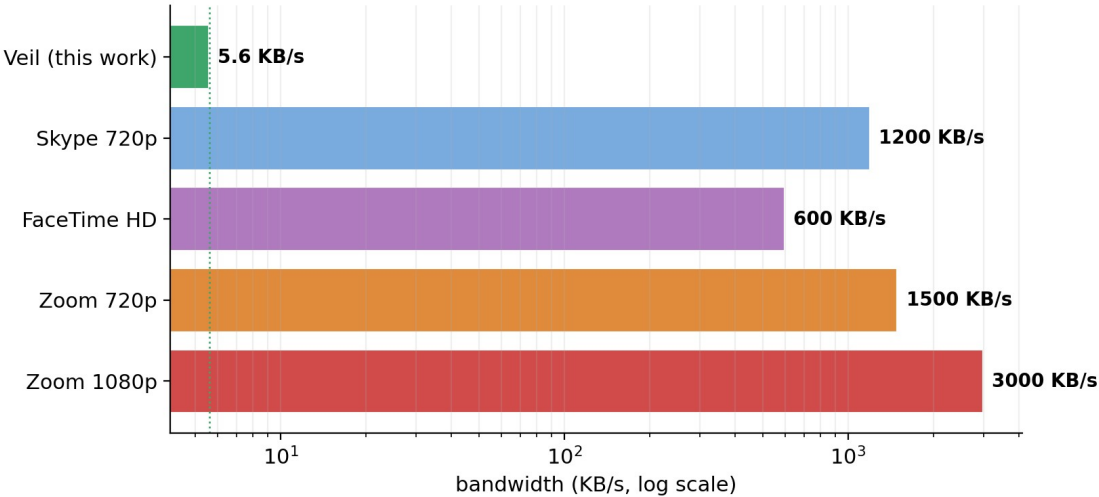
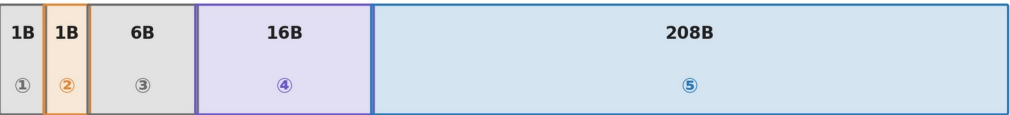


图 / Fig 9 · 带宽对比 / bandwidth comparison

encodePacket() · 233 B per frame · @ 24 Hz → 5.6 KB/s



① sender id (relay) · ② action idx · ③ header (reserved) · ④ head pose quaternion (xyzw) · ⑤ 52 × float32 ARKit blendshapes

Field widths drawn at √(size) scale so the 1-byte fields stay legible.

The actual 208 B blendshape payload is ≈ 89 % of the packet — header overhead is < 11 %.

图 / Fig 10 · 封包结构 / packet anatomy

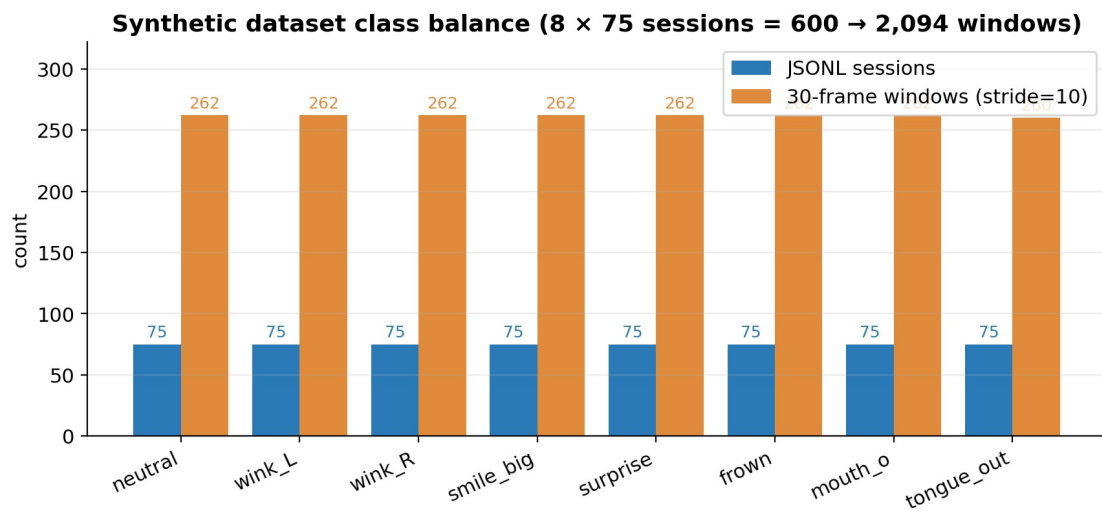


图 / Fig 11 · 数据集平衡 / dataset class balance